

VLSI INTERNSHIP – WEEK 1

Verilog HDL Codes for Digital Logic Circuits

Submitted By

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Electronics & Communication Engineering

Tools Used

- Xilinx Vivado

- CircuitVerse

Internship Duration

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1. AND Gate Verilog Code

```
module and_gate(  
    input A,  
    input B,  
    output Y  
);  
    assign Y = A & B;  
endmodule
```

Truth Table:

A	B	Y
---	---	---

0	0	0
---	---	---

0	1	0
---	---	---

1	0	0
---	---	---

1	1	1
---	---	---

2. OR Gate Verilog Code

```
module or_gate(
```

```
input A,
```

```
input B,
```

```
output Y
```

```
);
```

```
assign Y = A | B;
```

```
endmodule
```

Truth Table:

A	B	Y
---	---	---

0	0	0
---	---	---

0	1	1
---	---	---

1	0	1
---	---	---

1	1	1
---	---	---

3. NOT Gate Verilog Code

```
module not_gate(  
    input A,  
    output Y  
);  
    assign Y = ~A;  
endmodule
```

Truth Table:

A	Y
---	---

0	1
---	---

1	0
---	---

4. NAND Gate Verilog Code

```
module nand_gate(  
    input A,  
    input B,  
    output Y  
);  
    assign Y = ~(A & B);  
endmodule
```

Truth Table:

A	B	Y
---	---	---

0	0	1
---	---	---

0	1	1
---	---	---

1	0	1
---	---	---

1	1	0
---	---	---

5. NOR Gate Verilog Code

```
module nor_gate(  
    input A,  
    input B,  
    output Y  
);  
    assign Y = ~(A | B);  
endmodule
```

Truth Table:

A	B	Y
---	---	---

0	0	1
---	---	---

0	1	0
---	---	---

1	0	0
---	---	---

1	1	0
---	---	---

6. XOR Gate Verilog Code

```
module xor_gate(  
    input A,  
    input B,  
    output Y  
);  
    assign Y = A ^ B;  
endmodule
```

Truth Table:

A	B	Y
---	---	---

0	0	0
---	---	---

0	1	1
---	---	---

1	0	1
---	---	---

1	1	0
---	---	---

7. XNOR Gate Verilog Code

```
module xnor_gate(  
    input A,  
    input B,  
    output Y
```

```
);  
assign Y = ~(A ^ B);  
endmodule
```

Truth Table:

A	B	Y
---	---	---

0	0	1
---	---	---

0	1	0
---	---	---

1	0	0
---	---	---

1	1	1
---	---	---

8. Half Adder Verilog Code

```
module half_adder(  
    input A,  
    input B,  
    output SUM,  
    output CARRY  
);  
    assign SUM = A ^ B;  
    assign CARRY = A & B;  
endmodule
```

Half Adder Truth Table:

A	B	SUM	CARRY
---	---	-----	-------

0	0	0	0
---	---	---	---

0	1	1	0
---	---	---	---

1	0	1	0
---	---	---	---

1	1	0	1
---	---	---	---

9. Conclusion

Successfully implemented and verified:

- AND Gate
- OR Gate
- NOT Gate
- NAND Gate
- NOR Gate
- XOR Gate
- XNOR Gate
- Half Adder

using Verilog HDL in Xilinx Vivado and verified outputs through simulation waveforms and truth tables.